Lab Worksheet - Exploratory Data Analysis in airquality Dataset

```r
# load libraries:
library(tidyverse) # Recall that tidyverse contains dplyr, ggplot2, and many other useful packages.

# Load `airquality` dataset:
data("airquality")
airquality |>
    head()
```

```
  Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
6    28      NA 14.9   66     5   6
```

Q1: Render this quarto file to a pdf. To do so, you can use the hotkey `Ctrl + Shift + K` or `Cmd + Shift + K`. Then, you will see a command appear in the Positron Terminal. Once the command finishes running, you will see a pdf file appear in the same directory as this file and a VIEWER window will display the pdf on the right pane of Positron. Here you should see the rendered contents of this file.

Q2: Convert the `Month` column to a factor with it's levels labeled as month names (e.g. "May", "June", etc.).

```r
# your code here
airquality <- airquality |>
    mutate(Month = factor(Month, levels = 5:9, labels = c("May", "June", "July", "August", "September")))

airquality |>
    head()
```

```
  Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67   May   1
2    36     118  8.0   72   May   2
3    12     149 12.6   74   May   3
4    18     313 11.5   62   May   4
5    NA      NA 14.3   56   May   5
6    28      NA 14.9   66   May   6
```

Q3: Create a new column `Temp_C` that converts the temperature from Fahrenheit to Celsius.

```
# your code here
airquality <- airquality |>
    mutate(Temp_C = (Temp - 32) * 5/9)

airquality |>
    head()
```

```
  Ozone Solar.R Wind Temp Month Day   Temp_C
1    41     190  7.4   67   May   1 19.44444
2    36     118  8.0   72   May   2 22.22222
3    12     149 12.6   74   May   3 23.33333
4    18     313 11.5   62   May   4 16.66667
5    NA      NA 14.3   56   May   5 13.33333
6    28      NA 14.9   66   May   6 18.88889
```

Q4: Rename the `Wind` column to `Wind_mph` and the Temp column to `Temp_F`.

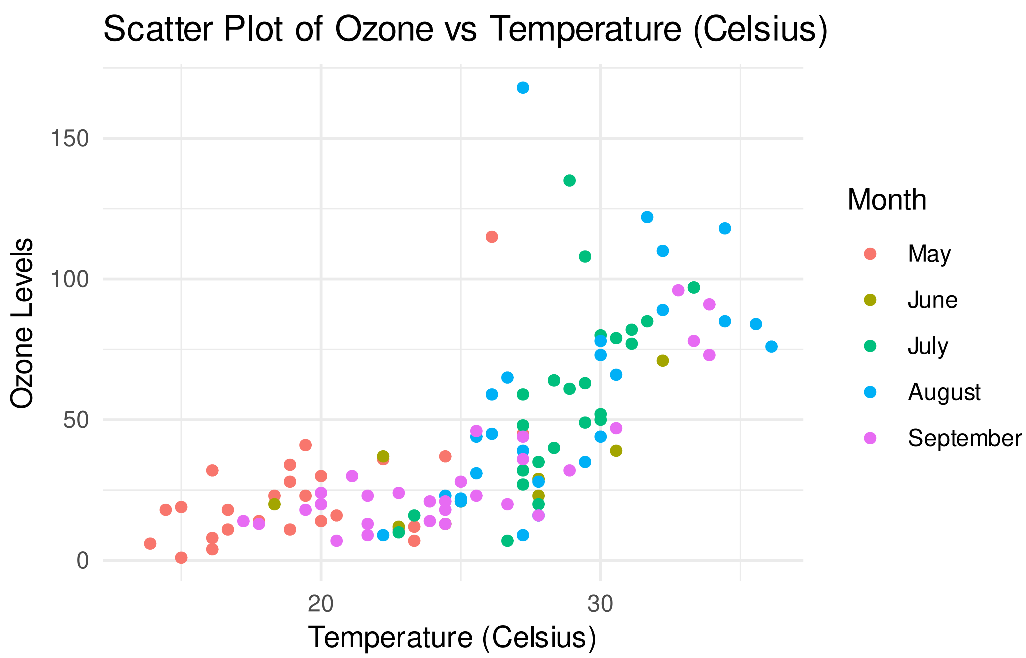
```
# your code here
airquality <- airquality |>
    rename(Wind_mph = Wind, Temp_F = Temp)

# or:
# airquality <- airquality |>
#     mutate(Temp_F = Temp, Wind_mph = Wind) |>
#     select(-Temp, -Wind)

airquality |>
    head()
```

```
  Ozone Solar.R Wind_mph Temp_F Month Day   Temp_C
1    41     190      7.4     67   May   1 19.44444
2    36     118      8.0     72   May   2 22.22222
3    12     149     12.6     74   May   3 23.33333
4    18     313     11.5     62   May   4 16.66667
5    NA      NA     14.3     56   May   5 13.33333
6    28      NA     14.9     66   May   6 18.88889
```

Q5: Create a scatter plot of `Ozone` vs `Temp_C`, colored by `Month`. Add appropriate axis labels and a title.

```r
# your code here
airquality |>
    ggplot(aes(x = Temp_C, y = Ozone, color = Month)) +
    geom_point() +
    labs(title = "Scatter Plot of Ozone vs Temperature (Celsius)",
         x = "Temperature (Celsius)",
         y = "Ozone Levels") +
    theme_minimal()
```

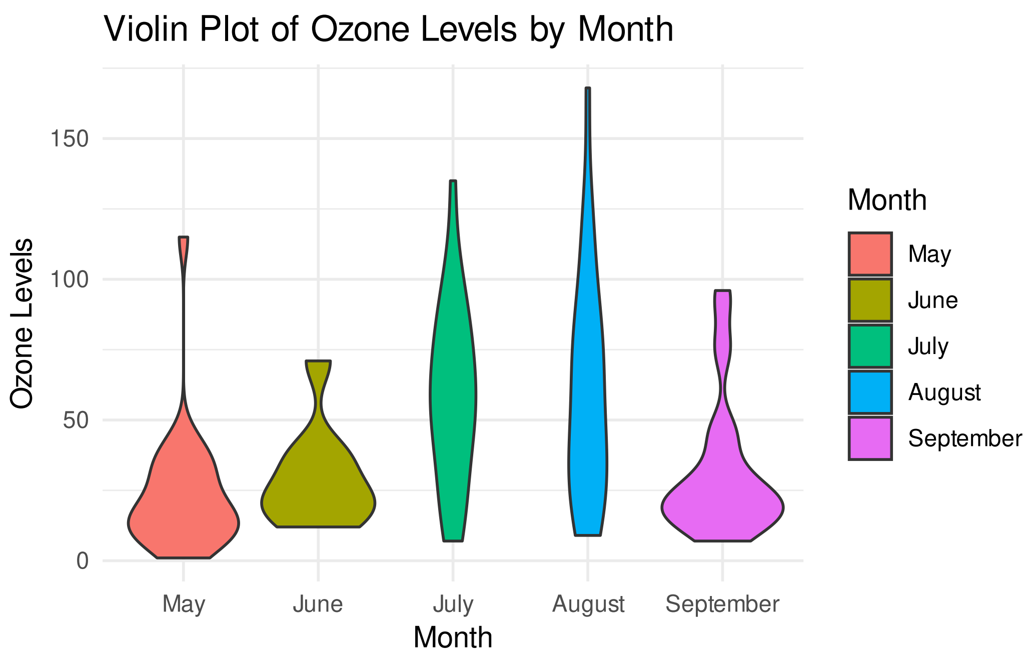
```
Warning: Removed 37 rows containing missing values or values outside the scale
range
(`geom_point()`).
```



Q6: Create a violin plot of `Ozone` levels for each month. Add appropriate axis labels and a title.

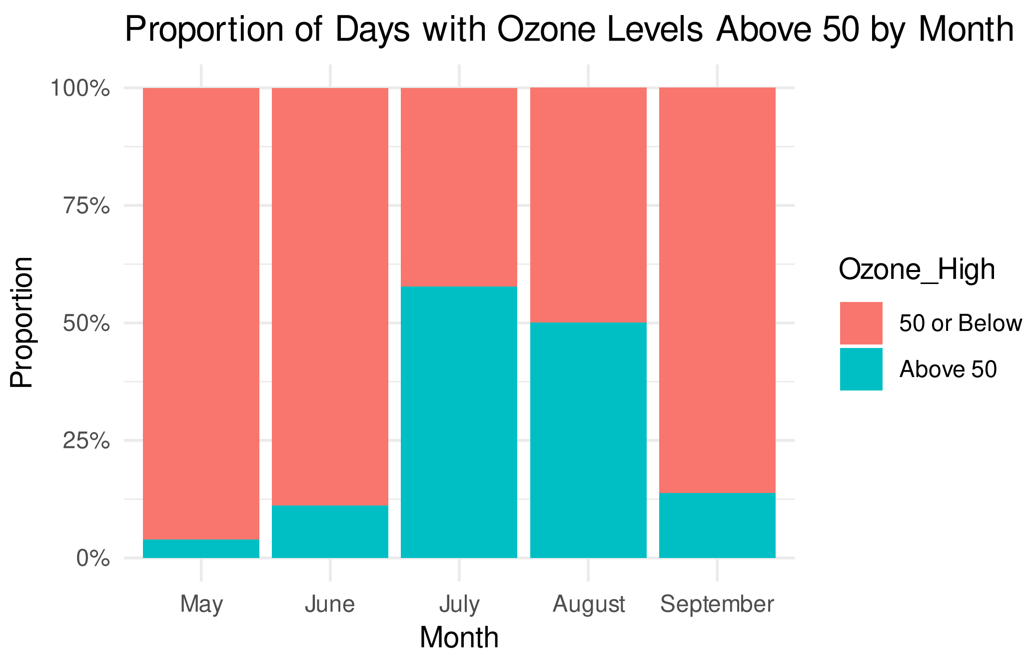
```
# your code here
airquality |>
    ggplot(aes(x = Month, y = Ozone, fill = Month)) +
    geom_violin() +
    labs(title = "Violin Plot of Ozone Levels by Month",
         x = "Month",
         y = "Ozone Levels") +
    theme_minimal()
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```

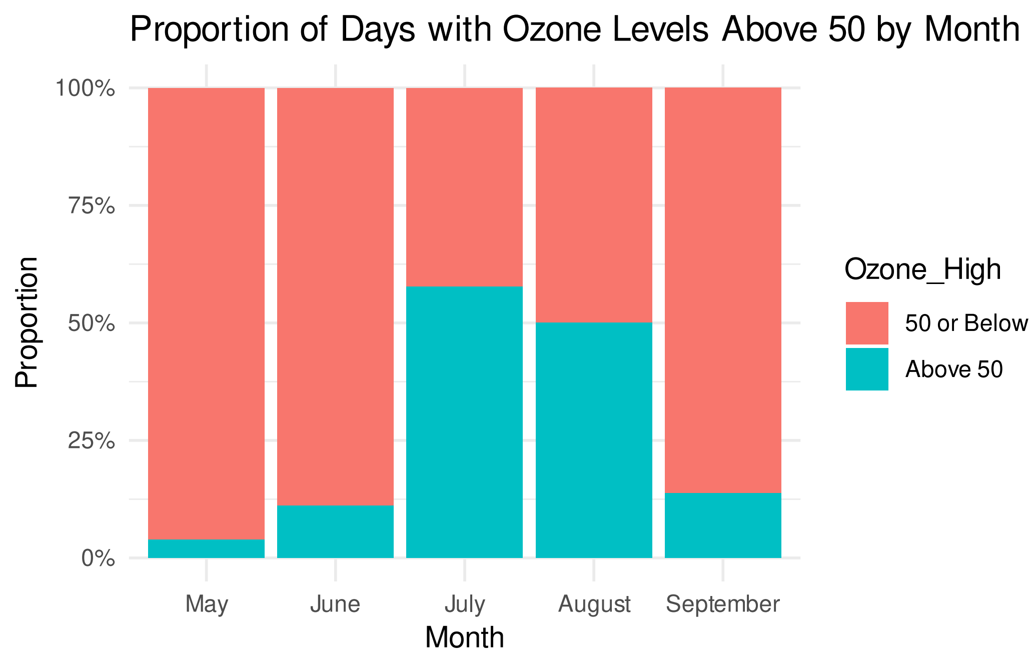


Q7: Create a segmented bar chart of the proportion of days with `Ozone` levels above 50 by `Month`. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
airquality |>
    drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
    mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
    group_by(Month, Ozone_High) |>
    summarise(Count = n(), .groups = 'drop') |>
    group_by(Month) |>
    mutate(Proportion = Count / sum(Count)) |>
    ggplot(aes(x = Month, y = Proportion, fill = Ozone_High)) +
    geom_bar(stat = "identity", position = "fill") +
    labs(title = "Proportion of Days with Ozone Levels Above 50 by Month",
         x = "Month",
         y = "Proportion") +
    scale_y_continuous(labels = scales::percent) +
    theme_minimal()
```

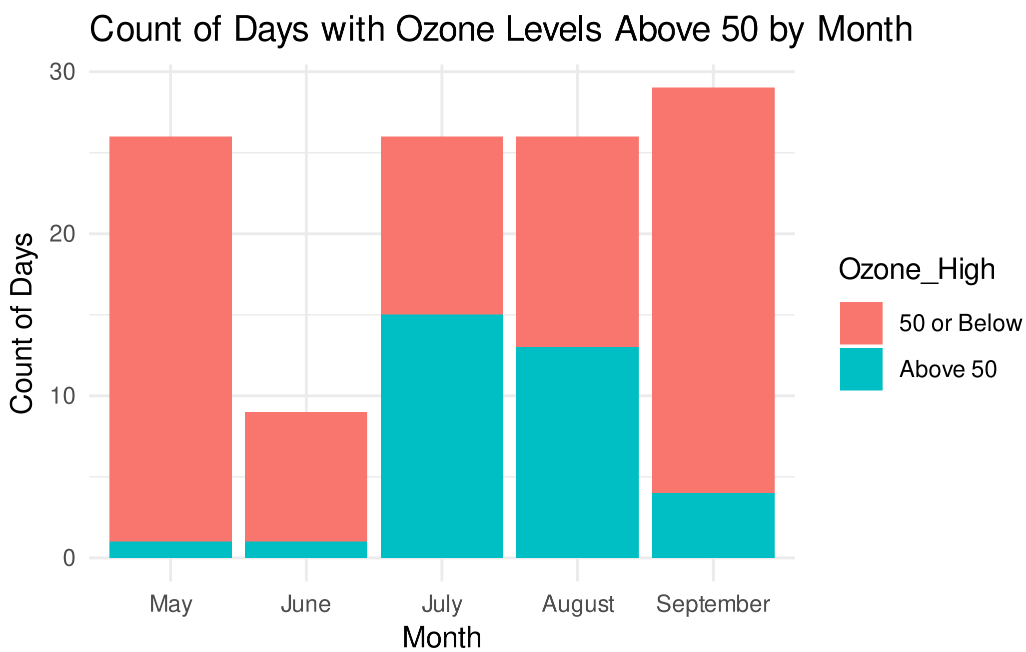



```
# or simply:
airquality |>
    drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
    mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
    ggplot(aes(x = Month, fill = Ozone_High)) + # count is the default
statistic in geom_bar, however we fix with position = "fill"
    geom_bar(position = "fill") + # position = "fill" gives proportions
instead
    labs(title = "Proportion of Days with Ozone Levels Above 50 by Month",
         x = "Month",
         y = "Proportion") +
    scale_y_continuous(labels = scales::percent) +
    theme_minimal()
```



Q8: Create a stacked bar chart showing the count of days available for each month. Fill bar colors by `Ozone` levels above or below 50 for each month. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
airquality |>
    drop_na(Ozone) |> # or filter(!is.na(Ozone)) |>
    mutate(Ozone_High = ifelse(Ozone > 50, "Above 50", "50 or Below")) |>
    ggplot(aes(x = Month, fill = Ozone_High)) +
    geom_bar() +
    labs(title = "Count of Days with Ozone Levels Above 50 by Month",
         x = "Month",
         y = "Count of Days") +
    theme_minimal()
```



Q9: What is the utility of using a segmented bar chart vs a regular bar chart in this context? Which do you think is more informative and why?

…

Q10: Create a summary table of the number and proportion of NA values in each column.

```
# your code here
airquality |>
    summarize(
        mising_Ozone = sum(is.na(Ozone)),
        prop_missing_Ozone = mean(is.na(Ozone)),
        missing_SolarR = sum(is.na(Solar.R)),
        prop_missing_SolarR = mean(is.na(Solar.R)),
        missing_Wind = sum(is.na(Wind_mph)),
        prop_missing_Wind = mean(is.na(Wind_mph)),
        missing_Temp = sum(is.na(Temp_F)),
        prop_missing_Temp = mean(is.na(Temp_F)),
        missing_Month = sum(is.na(Month)),
        prop_missing_Month = mean(is.na(Month)),
        missing_Day = sum(is.na(Day)),
        prop_missing_Day = mean(is.na(Day))
```

```
  ) |>
  pivot_longer(everything(), names_to = "Metric", values_to = "Value")
```

```
# A tibble: 12 × 2
   Metric                 Value
   <chr>                  <dbl>
 1 mising_Ozone        37
 2 prop_missing_Ozone   0.242
 3 missing_SolarR       7
 4 prop_missing_SolarR  0.0458
 5 missing_Wind         0
 6 prop_missing_Wind    0
 7 missing_Temp         0
 8 prop_missing_Temp    0
 9 missing_Month        0
10 prop_missing_Month   0
11 missing_Day          0
12 prop_missing_Day     0
```

```
airquality |>
  summarize(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Column", values_to = "Num_NA")
```

```
# A tibble: 7 × 2
  Column   Num_NA
  <chr>     <int>
1 Ozone        37
2 Solar.R       7
3 Wind_mph      0
4 Temp_F        0
5 Month         0
6 Day           0
7 Temp_C        0
```

```
airquality |>
  summarize(across(everything(), ~ mean(is.na(.)))) |>
  pivot_longer(everything(), names_to = "Column", values_to = "Prop_NA")
```

```
# A tibble: 7 × 2
  Column   Prop_NA
  <chr>      <dbl>
```

```
1 Ozone     0.242
2 Solar.R   0.0458
3 Wind_mph  0
4 Temp_F    0
5 Month     0
6 Day       0
7 Temp_C    0
```